

Invariance

STAT 220B – Lecture 11

Stats 220B

Where we left off

Lectures 9 and 10 finished the minimax chapter with the Stein phenomenon: for $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ with $p \geq 3$, the MLE $\delta^0 = \mathbf{X}$ is minimax but inadmissible.

Two lingering questions hang over that story:

Why is the MLE minimax in the first place? Lecture 8 used the sequence-of-priors trick (Theorem 18) with $\mathcal{N}(0, k)$ priors. The MLE just happened to have constant risk equal to the limiting Bayes risk. Why does this *always* work for location problems? Is there a structural reason, not just an accidental computation?

Why is Stein's improvement possible? If something forces the MLE to be minimax for any sample size and any p , what is that thing, and why is it silent about admissibility?

Today we introduce the structural principle behind both questions: **invariance**, the fourth selection principle from Lecture 1 (alongside Bayes, admissibility, minimax). A problem with symmetry has only a few estimators that respect the symmetry. Among those, the best one is usually minimax. Whether it is also admissible is a separate question (and the Stein phenomenon is the answer at $p \geq 3$).

Reading: Berger, Chapter 6 (sections 6.1 through 6.5). Today is dense; we sweep through the entire chapter. The supplementary notes will fill in technical details (group theory background, Haar measure, formal Hunt-Stein proof).

Why invariance? (Berger §6.1)

A concrete motivation (Berger Example 1)

A particle's decay time X is exponentially distributed with mean $\theta > 0$, density

$$f(x \mid \theta) = \theta^{-1} e^{-x/\theta}, \quad x > 0.$$

Estimate θ from one observation X , under loss $L(\theta, a) = (1 - a/\theta)^2$.

A statistician in Tokyo measures time in seconds and proposes the rule $\delta_0(x)$.

A statistician in Boston measures the same physical process in minutes, observing $Y = X/60$. With $\eta = \theta/60$ (mean decay in minutes), $Y \sim \text{Exp}(\eta)$. Writing $a^* = a/60$, the loss is

$$L(\theta, a) = \left(1 - \frac{a}{\theta}\right)^2 = \left(1 - \frac{a/60}{\theta/60}\right)^2 = \left(1 - \frac{a^*}{\eta}\right)^2 = L(\eta, a^*).$$

The *structure* of the Boston problem (density family, parameter space, loss) is identical to Tokyo's. A reasonable demand: use the same decision rule.

What “same rule” forces [board example]

Tokyo uses δ_0 on X . Boston uses δ_0 on Y (same rule). Converting Boston's answer back to seconds: Tokyo should get $60 \cdot \delta_0(Y) = 60 \cdot \delta_0(X/60)$. But Tokyo *also* directly applies $\delta_0(X)$. Consistency:

$$\delta_0(X) = 60 \cdot \delta_0(X/60).$$

Replace 60 by any $c > 0$. For every $c > 0$:

$$\delta_0(X) = c \cdot \delta_0(X/c).$$

Setting $c = 1/X$: $\delta_0(X) = X \cdot \delta_0(1)$. So the *only* rules consistent with our reasoning are

$\delta_0(x) = Kx \quad \text{for some constant } K > 0.$

The infinite-dimensional space of decision rules has collapsed to a one-parameter family, because we demanded the rule be unit-blind.

Two principles, one consequence

Berger distinguishes two intuitions in the example above:

Principle of rational invariance. The action taken in a decision problem should not depend on the unit of measurement or other arbitrary incidentals.

Invariance principle. If two decision problems have the same formal structure (same $\mathcal{X}$, Θ , family $\mathcal{F}$, loss L), the same decision rule should be used in each.

These two together say: when a transformation maps a problem to itself, the rule should respect that mapping. The resulting class of rules is the **invariant** (or **equivariant**) rules.

A caveat. Invariance ignores prior information. If we had strong prior beliefs about θ , invariance would override them by saying “use the rule that ignores units,” even if our prior reflects units. So invariance is the principle of choice when no prior information is at hand. That is exactly the setting where minimax and noninformative-Bayes analyses live.

Finishing the example

For the exponential decay problem with $\delta_K(x) = Kx$ and loss $L(\theta, a) = (1 - a/\theta)^2$:

$$R(\theta, \delta_K) = \mathbb{E}_\theta \left[\left(1 - \frac{KX}{\theta} \right)^2 \right] = \mathbb{E}_\theta \left[\left(1 - K \cdot \frac{X}{\theta} \right)^2 \right].$$

Since $X/\theta \sim \text{Exp}(1)$ has mean 1 and variance 1:

$$R(\theta, \delta_K) = (1 - K)^2 + K^2 = 1 - 2K + 2K^2, \quad \text{constant in } \theta.$$

Minimizing: $K^* = 1/2$. The best equivariant rule is $\delta^*(x) = x/2$, with constant risk $1/2$.

Three features to note that will recur throughout today:

- ▶ **No prior needed.** The argument used only the model and the loss.
- ▶ **No minimax computation needed.** The risk was constant, so the maximum is the value.
- ▶ **Equivariance reduced an infinite-dimensional space of rules to a one-dimensional family.** That is the value proposition of invariance: drastic reduction of the rule space.

This is what we now make formal.

The group-theoretic setup (Berger §6.2)

Three groups

An invariant decision problem requires three coordinated group actions. Each acts on a different space.

The three groups

A decision problem is **invariant under a group** $\mathcal{G}$ if there are three groups

$$\mathcal{G} = \{g\}, \quad \bar{\mathcal{G}} = \{\bar{g}\}, \quad \tilde{\mathcal{G}} = \{\tilde{g}\},$$

acting respectively on the sample space $\mathcal{X}$, the parameter space Θ , and the action space $\mathcal{A}$, satisfying:

1. **Model invariance.** $X \sim P_\theta \Rightarrow gX \sim P_{\bar{g}\theta}$ for all $g \in \mathcal{G}$.
2. **Loss invariance.** $L(\bar{g}\theta, \tilde{g}a) = L(\theta, a)$ for all θ, a, g .
3. **Compatibility.** The mappings $g \mapsto \bar{g}$ and $g \mapsto \tilde{g}$ are group homomorphisms.

In a location problem (Berger Example 1), all three groups are the translation group

$\mathcal{G} = \{g_c(x) = x + c : c \in \mathbb{R}\}$, acting identically on each space. But $\bar{\mathcal{G}}$ and $\tilde{\mathcal{G}}$ need not equal $\mathcal{G}$ in general.

Example 1: location families

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x - \theta)$ for some density f , $\theta \in \mathbb{R}$. Estimate θ under loss $L(\theta - a)$ (depending on θ and a only through the difference).

Take $\mathcal{G} = \bar{\mathcal{G}} = \tilde{\mathcal{G}} = (\mathbb{R}, +)$, with $g_c(x) = x + c$ for each space.

Check the three conditions.

1. **Model.** If $X_i = \theta + \epsilon_i$ with ϵ_i i.i.d. from f , then $X_i + c = (\theta + c) + \epsilon_i$. So the shifted data has the original error structure with new parameter $\theta + c$. Model invariant.
2. **Loss.** $L((\theta + c) - (a + c)) = L(\theta - a)$. Loss invariant.
3. **Compatibility.** The map $c \mapsto c$ (identity) is trivially a homomorphism.

This is the prototype. Every translation-invariant problem fits this template.

Example 2: scale families

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} (1/\sigma)f(x/\sigma)$ for $\sigma > 0$. Estimate σ under loss $L(\sigma, a) = W(a/\sigma)$ for some function W (scale-invariant loss).

Take $\mathcal{G} = \bar{\mathcal{G}} = \tilde{\mathcal{G}} = (\mathbb{R}_+, \cdot)$, the multiplicative group of positive reals, with $g_c(x) = cx$.

The three conditions are checked similarly: scaling the data by c corresponds to scaling σ by c , and the loss depends only on the ratio a/σ .

General template. Whenever the parameter family is closed under some group action and the loss respects the same action, the problem is invariant.

Invariant decision rules (Berger §6.2)

Equivariance

Definition

A decision rule $\delta : \mathcal{X} \rightarrow \mathcal{A}$ is **equivariant** (or **invariant**, the terms are used interchangeably for nonrandomized rules) under $\mathcal{G}$ if

$$\delta(gx) = \tilde{g}\delta(x) \quad \text{for all } x \in \mathcal{X}, g \in \mathcal{G}.$$

In words: the rule commutes with the group action. Shift the data, and the action shifts the same way.

For the location problem with $\mathcal{G} = (\mathbb{R}, +)$: equivariance says $\delta(x + c) = \delta(x) + c$ for all c .

For the scale problem with $\mathcal{G} = (\mathbb{R}_+, \cdot)$: equivariance says $\delta(cx) = c\delta(x)$ for all $c > 0$.

Why this is the right concept. Equivariance is exactly the condition that the rule's behavior is intrinsic to the data, not to the choice of coordinate system. Two analysts using different units (different c 's) will report estimates that agree once translated to a common frame.

The Invariance Principle

Principle of Invariance (Berger §6.2)

If a decision problem is invariant under a group $\mathcal{G}$, **only equivariant rules should be considered.**

This is the fourth selection principle. Like the others (Bayes, minimax, admissibility), it is a normative claim, not a theorem.

Why it is reasonable.

- ▶ **Coordinate freedom.** No legitimate inference should depend on an arbitrary choice of origin or scale.
- ▶ **Logical consistency.** If two problems are “the same up to relabeling,” they should have the same solution up to relabeling.
- ▶ **Computational reduction.** The equivariant family is usually much smaller than the full rule space.

Why it can be misleading. The same data can be invariant under multiple groups, and the “best invariant rule” depends on which group. Also, as we’ll see, invariance can conflict with admissibility. So invariance is a guide, not a guarantee.

Risk of equivariant rules

Theorem 1 (Berger §6.2)

If δ is equivariant under $\mathcal{G}$, then $R(\theta, \delta) = R(\bar{g}\theta, \delta)$ for every $\bar{g} \in \bar{\mathcal{G}}$.

Proof. Using model invariance and loss invariance:

$$\begin{aligned} R(\bar{g}\theta, \delta) &= \mathbb{E}_{P_{\bar{g}\theta}}[L(\bar{g}\theta, \delta(X))] \\ &= \mathbb{E}_{P_{\theta}}[L(\bar{g}\theta, \delta(gX))] && \text{(model invariance)} \\ &= \mathbb{E}_{P_{\theta}}[L(\bar{g}\theta, \tilde{g}\delta(X))] && \text{(equivariance)} \\ &= \mathbb{E}_{P_{\theta}}[L(\theta, \delta(X))] = R(\theta, \delta) && \text{(loss invariance).} \end{aligned}$$

Consequence. The risk of an equivariant rule is **constant on orbits of $\bar{\mathcal{G}}$ in Θ** .

For the location problem ($\bar{\mathcal{G}}$ acts transitively on $\Theta = \mathbb{R}$), the orbit is all of Θ . So **equivariant rules have constant risk in θ** . The minimax problem becomes trivial: minimize the constant.

Maximal invariants and reduction (Berger §6.3)

The orbit structure

The group $\mathcal{G}$ partitions $\mathcal{X}$ into **orbits**: $\mathcal{O}_x = \{gx : g \in \mathcal{G}\}$.

Similarly $\bar{\mathcal{G}}$ partitions Θ into orbits, and $\tilde{\mathcal{G}}$ partitions $\mathcal{A}$ into orbits.

A function $T : \mathcal{X} \rightarrow \mathcal{T}$ is **invariant under $\mathcal{G}$** if $T(gx) = T(x)$ for all g, x . It is **maximal invariant** if it is invariant AND distinguishes orbits: $T(x_1) = T(x_2) \Rightarrow x_2 = gx_1$ for some g .

Maximal invariants are to invariance what sufficient statistics are to the likelihood. They are the “data-reducing” objects of the theory.

Supplementary notes (to be posted) work out: (1) why every invariant function factors through a maximal invariant, (2) how to construct maximal invariants on Θ from those on $\mathcal{X}$, (3) the formal statement of the reduction theorem.

Example: location maximal invariant

For $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ under translation $g_c(x) = (x_1 + c, \dots, x_n + c)$:

The orbits are sets of the form $\{(x_1 + c, x_2 + c, \dots, x_n + c) : c \in \mathbb{R}\}$.

A natural invariant: any pairwise difference $X_i - X_j$. A more principled choice: the vector of differences from X_n ,

$$T(X) = (X_1 - X_n, X_2 - X_n, \dots, X_{n-1} - X_n).$$

Claim: T is maximal invariant.

Proof. $T(gx) = T(x)$ since shifting all coordinates by c cancels in differences. So T is invariant. Conversely, if $T(x) = T(y)$, then $x_i - x_n = y_i - y_n$ for all i , so $x_i = y_i + (x_n - y_n)$. Setting $c = x_n - y_n$ gives $x = g_c y$, so x and y are in the same orbit. $\square$

The maximal invariant has dimension $n - 1$. Loosely: equivariance removes one dimension (the location dimension) from the data.

The reduction theorem (informal)

Reduction Principle

The distribution of a maximal invariant statistic $T(X)$ depends on θ only through a **maximal invariant on Θ** .

For the location problem: $T(X) = (X_1 - X_n, \dots, X_{n-1} - X_n)$. The distribution of $T(X)$ is the same for every θ , since shifting θ shifts all X_i by the same amount.

Consequence. $T(X)$ has a distribution that does not depend on θ . It carries no information about θ in the classical sense.

So how can the data tell us anything about θ ? **Through X_n (or any other choice of origin).** Conditional on $T(X)$, the conditional distribution of X_n given $T(X)$ does depend on θ , and that is where the inference lives.

The Pitman estimator (Berger §6.4)

Best equivariant rule: location families

For the location problem, every equivariant rule has the form

$$\delta(X) = X_n + h(T(X))$$

for some function $h : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$, where $T(X) = (X_1 - X_n, \dots, X_{n-1} - X_n)$ is the maximal invariant.

(Reason: $\delta(X) - X_n = \delta(X - X_n \cdot \mathbf{1})$ by equivariance, which depends only on differences.)

Among all such h , we want the one minimizing the constant risk $R(\theta, \delta)$.

For squared-error loss $L(\theta - a) = (\theta - a)^2$:

$$R(0, \delta) = \mathbb{E}_0[(X_n + h(T) - 0)^2] = \mathbb{E}_0[(X_n + h(T))^2].$$

Fix $T = t$ and minimize over $h(t)$. The inner conditional expectation is minimized at

$$h^*(t) = -\mathbb{E}_0[X_n \mid T = t].$$

The Pitman estimator

Pitman Estimator (Berger §6.4)

For a location family with squared-error loss, the **best (minimum-risk) equivariant estimator** is

$$\delta^*(X) = X_n + h^*(T(X)) = X_n - \mathbb{E}_0[X_n \mid T(X)].$$

Equivalently, using the integral form (and assuming the density f is given):

$$\delta^*(X) = \frac{\int_{-\infty}^{\infty} u \prod_{i=1}^n f(X_i - u) du}{\int_{-\infty}^{\infty} \prod_{i=1}^n f(X_i - u) du}.$$

Reading the integral form. The numerator is the posterior mean of u under the formal prior $\pi(u) = 1$, with u playing the role of θ . The denominator normalizes. So:

The Pitman estimator is the generalized Bayes rule under the uniform improper prior.

This is the cleanest illustration of the deep connection between invariance and generalized Bayes rules with right-invariant Haar priors. (Theorem 6 in Berger.)

Supplementary notes: full proof that the Pitman estimator equals the generalized Bayes rule under the right-invariant Haar measure on the group. For the location group, this reduces to Lebesgue. For the scale group, it gives $\pi(\sigma) = 1/\sigma$.

Example: normal mean

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, squared-error loss.

The density $f(x - \theta) = (2\pi)^{-1/2} \exp\{-(x - \theta)^2/2\}$. Then

$$\prod_{i=1}^n f(X_i - u) = (2\pi)^{-n/2} \exp\left\{-\frac{1}{2} \sum_{i=1}^n (X_i - u)^2\right\}.$$

Completing the square in u :

$$\sum (X_i - u)^2 = n(u - \bar{X})^2 + \sum (X_i - \bar{X})^2.$$

The second term is constant in u . The integral becomes:

$$\delta^*(X) = \frac{\int u e^{-n(u - \bar{X})^2/2} du}{\int e^{-n(u - \bar{X})^2/2} du} = \bar{X}.$$

So for normal location, the Pitman estimator is the sample mean, $\delta^*(X) = \bar{X}$.

We have established that $\bar{X}$ is the **best location-equivariant estimator** for the normal location problem.

The conflict, foreshadowed

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$ with $\theta \in \mathbb{R}$ (1-dimensional), $\bar{X}$ is best equivariant. It is also minimax (Lecture 8) and admissible (a deeper result, Berger §8.7).

For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ with $\theta \in \mathbb{R}^p$ and $p \geq 3$, the analogous calculation shows that $\bar{\mathbf{X}} = \mathbf{X}$ is best equivariant under translation $g_c \mathbf{x} = \mathbf{x} + c$.

But Lecture 10 proved that $\mathbf{X}$ is **inadmissible** for $p \geq 3$. The James-Stein estimator δ^{JS} strictly dominates it.

Invariance vs admissibility

For $\mathcal{N}_p(\theta, I_p)$ with $p \geq 3$: the best location-equivariant estimator is inadmissible.

The James-Stein estimator that dominates it is **not equivariant**: it shrinks toward an arbitrary point (the origin), which breaks translation symmetry.

This is one of the deepest tensions in classical decision theory. We return to it in the closing remarks.

Hunt-Stein theorem (Berger §6.5)

The minimax connection

We have seen: equivariant rules under a transitive group action have constant risk. So among equivariant rules, **the best equivariant rule is automatically the minimax equivariant rule.**

Question. Is the best equivariant rule also minimax in the full sense (minimizing $\sup_{\theta} R$ over **all** rules, not just equivariant ones)?

Answer (Hunt-Stein). Under group-theoretic regularity (amenability), yes. The best invariant rule is minimax.

This gives a powerful method for finding minimax rules in invariant problems: just compute the best equivariant rule, which is usually much easier than a direct minimax search.

Amenability (informal)

A group $\mathcal{G}$ is **amenable** if it admits an asymptotically invariant sequence of probability measures (a “Folner sequence”). Equivalently, the group has approximately-invariant averaging operators.

Examples of amenable groups.

- ▶ $(\mathbb{R}^n, +)$: the translation group, with averages over balls of growing radius.
- ▶ $(\mathbb{R}_+, \cdot)$: the scale group, via logarithmic transformation to $(\mathbb{R}, +)$.
- ▶ Compact groups (orthogonal group $O(n)$, etc.).
- ▶ Solvable groups in general.

Non-amenable groups include the general linear group $GL(n)$ for $n \geq 2$ and free groups. For these, Hunt-Stein can fail and the best equivariant rule may not be minimax.

For most statistical problems of interest (location, scale, location-scale), the relevant group is amenable, and Hunt-Stein applies.

Hunt-Stein theorem (statement)

Hunt-Stein (Berger Thm.~6.6.10)

Suppose:

1. The decision problem is invariant under an amenable group $\mathcal{G}$.
2. The loss is bounded below.
3. The risk function $R(\theta, \delta)$ is suitably regular (continuous in θ).

Let δ^* be the best equivariant rule. Then δ^* is **minimax**:

$$\sup_{\theta} R(\theta, \delta^*) = \inf_{\delta} \sup_{\theta} R(\theta, \delta).$$

Proof idea (informal). Given any rule δ_0 with $\sup_{\theta} R(\theta, \delta_0) < \infty$, construct an equivariant version

$$\delta_{\bar{\mathcal{G}}}(x) = \int_{\mathcal{G}} \tilde{g}^{-1} \delta_0(gx) d\nu(g)$$

by averaging against an approximately-invariant probability measure ν on $\mathcal{G}$ (this is where amenability is used). The averaged rule has $\sup_{\theta} R(\theta, \delta_{\bar{\mathcal{G}}}) \leq \sup_{\theta} R(\theta, \delta_0)$. The best equivariant rule is at least as good as the averaged version, so it dominates.

Supplementary notes: full proof of Hunt-Stein with Haar measure background, the Folner sequence construction, and the dominated-convergence step. Worked example for the location group on $\mathbb{R}^p$.

Application: minimax under translation

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x - \theta)$, $\theta \in \mathbb{R}$, squared-error loss, with f such that $\text{Var}_f < \infty$:

1. Translation group $(\mathbb{R}, +)$ is amenable.
2. Loss is bounded below (it equals $(\theta - a)^2 \geq 0$).
3. Pitman estimator is the best equivariant rule.

By Hunt-Stein, the Pitman estimator is minimax.

For the normal case: the Pitman estimator is $\bar{X}$. We recover the Lecture 8 result that $\bar{X}$ is minimax. But the invariance route was easier: no equalizer-Bayes argument needed.

For non-normal location families (e.g., Laplace, Cauchy, double-exponential), the Pitman estimator is generally not the MLE or the sample mean, but it is still minimax by the same argument.

Bayes, minimax, invariance: how they fit together

The general picture

Across Lectures 4-11, the four selection principles connect in a now-recognizable web:

	Best equivariant	Minimax
Connection to Bayes	Generalized Bayes under right-invariant Haar prior (when group is amenable)	Equalizer Bayes (Thm 17) or extended Bayes (Thm 18)
When they agree	By Hunt-Stein, when group is amenable	—
When they conflict	Stein phenomenon ($p \geq 3$): best equivariant is inadmissible	—

The Stein conflict. Hunt-Stein says best equivariant is minimax. Stein says best equivariant is inadmissible (for $\mathcal{N}_p$, $p \geq 3$). So minimax is inadmissible in that setting. The James-Stein estimator is admissible-improvable (by Baranchik), not minimax, not equivariant, but uniformly better than the best equivariant minimax rule.

Resolution: no resolution. These tensions are intrinsic to the theory. They are not bugs to be patched. They are features that tell us no single principle dominates.

Why this matters

Each principle answers a different question:

- ▶ **Bayes** answers: “What rule is optimal given my prior beliefs?”
- ▶ **Admissibility** answers: “What rules are not strictly worse than alternatives?”
- ▶ **Minimax** answers: “What rule protects against the worst case?”
- ▶ **Invariance** answers: “What rule respects the symmetries of the problem?”

These are different questions, and they have different answers in dimensions ≥ 3 . Practical statistical work picks among them based on context:

- ▶ **Few replications, want admissibility:** Bayesian shrinkage (empirical or fully Bayesian).
- ▶ **Many replications, want worst-case guarantees:** Minimax (often via equivariance when applicable).
- ▶ **Symmetric problem, want coordinate-free answer:** Equivariant rule.

The theoretical literature distinguishes these carefully because they distinguish themselves.

Summary

What we built

Group structure (\$\$\$6.2). An invariant problem has three groups $\mathcal{G}, \bar{\mathcal{G}}, \tilde{\mathcal{G}}$ acting on $\mathcal{X}, \Theta, \mathcal{A}$, satisfying model invariance, loss invariance, and compatibility.

Equivariance. A rule δ is equivariant if $\delta(gx) = \tilde{g}\delta(x)$. Equivariant rules have constant risk on orbits of $\bar{\mathcal{G}}$ in Θ .

Maximal invariants (\$\$\$6.3). A maximal invariant statistic $T(X)$ distinguishes orbits. Every equivariant rule has the form $\delta(X) = X_n + h(T(X))$ in the location case.

Pitman estimator (\$\$\$6.4). For location families with squared-error loss, the best equivariant estimator is $\delta^*(X) = X_n - \mathbb{E}_0[X_n \mid T(X)]$, which equals the generalized Bayes rule under uniform prior. For normal location, $\delta^* = \bar{X}$.

Hunt-Stein theorem (\$\$\$6.5). For amenable groups, the best equivariant rule is minimax. So invariance gives a route to minimax that is often easier than direct equalizer arguments.

The conflict. For $\mathcal{N}_p$ with $p \geq 3$, the best equivariant minimax rule ($\bar{X} = \mathbf{X}$) is inadmissible. The James-Stein estimator dominates but is not equivariant. **Selection principles disagree in higher dimensions.**

Step back: where invariance fits in the course

Today's material connects to every previous lecture in a specific way. Holding these connections in mind helps:

- ▶ **Lecture 1 (selection principles).** Invariance is the fourth of the four principles previewed there. We have now developed all four (Bayes, admissibility, minimax, invariance).
- ▶ **Lecture 3 (sufficiency).** Maximal invariants play, for equivariance, the role that sufficient statistics play for the likelihood. Both reduce the rule space.
- ▶ **Lecture 4 (Bayes rules).** The Pitman estimator is the generalized Bayes rule under the right-invariant Haar prior. Invariance and Bayes are not separate worlds.
- ▶ **Lecture 5 (admissibility of Bayes rules).** Generalized Bayes rules from improper priors lose admissibility protection. The Pitman estimator inherits this risk.
- ▶ **Lectures 6–8 (minimax).** Hunt-Stein says the best equivariant rule is minimax. We just got a non-trivial route to minimax that does not require equalizer-Bayes arguments.
- ▶ **Lectures 9–10 (Stein).** Best equivariant minimax is inadmissible for $\mathcal{N}_p$, $p \geq 3$. The James-Stein dominator is not equivariant. Invariance protects against everything except dimension.

One-sentence takeaway. Invariance is the principle that turns symmetries of the problem into restrictions on the rule space, often pinning down the minimax rule for free, but offering no guarantee of admissibility in higher dimensions.

Reading and final remarks

Berger, Chapter 6 (sections 6.1-6.5). The supplementary notes (to be posted) will give:

- ▶ Formal group-theoretic background (Haar measure, amenability).
- ▶ Proof sketches for the equivariance-Bayes connection (Berger Thm. 6).
- ▶ Hunt-Stein proof outline.

Exercises. Berger §6 Exercises 1, 4, 7, 11, 19 (Pitman for normal mean), 21 (Pitman for double exponential), 27 (Hunt-Stein for scale).

That is the course. Across the quarter we built four selection principles (Bayes, admissibility, minimax, invariance), the sufficiency machinery that supports them, the utility foundation underneath, and the Stein phenomenon that shows them disagreeing in dimension three and higher. The cheatsheet covers all of this.